

Marathon Petroleum Corporation

US Refining Giant + MPLX Midstream MLP — Largest Refiner by Capacity

PhD Due Diligence Report | BLRI-DD-MPC-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Marathon Petroleum: Scale + MPLX Midstream = Superior Integrated Downstream

Marathon Petroleum is the largest US petroleum refiner by capacity (3M barrels/day, 13 refineries) and controls MPLX LP — a large-cap midstream MLP with 25,000+ miles of pipeline. At \$350.875 (mkt cap \$98.534B, P/E 12.17x, EPS \$13.22), MPC is near its 52-week high (\$351.92) with the most powerful share of each refining dollar among the three refiners. MPC's control of MPLX provides: (1) fee-based midstream income immune to commodity cycles, and (2) a monetisation path via MPLX dropdown transactions that generate non-dilutive capital for MPC buybacks.

Source: moomoo OpenD [2026-08-13] · Marathon Petroleum 10-Q EDGAR · MPLX LP 2026 annual report

PART II — QUALITATIVE ANALYSIS

MPLX: The Midstream Engine That Funds MPC's Buyback Machine

MPC has repurchased 40%+ of its shares outstanding over the past 5 years — one of the most aggressive buyback programmes in the S&P 500. The mechanism: MPLX distributes \$3-4B annually to MPC in distributions, providing non-refining cash flow that funds buybacks even when crack spreads are weak. MPC's Galveston Bay refinery (630,000 B/day) is the second-largest US refinery, able to process heavy Canadian crude at \$15+ discount — the same crack spread premium advantage as Valero's Port Arthur.

Source: MPC MPLX distribution history · Galveston Bay refinery capacity · MPC buyback programme history

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$350.875 | Prev Close: \$348.25 | Mkt Cap: \$98.534B
EPS TTM: \$13.22 | Net Income: \$3.713B | Div Yield: 1.11%
52w: \$157.78–\$351.92 | Shares: 280.8M

Metric	Value	Source
Price (2026-08-13)	\$350.875	moomoo OpenD
Market Cap	\$98.534B	moomoo OpenD
EPS TTM	\$13.22	moomoo OpenD
Net Income TTM	\$3.713B	moomoo OpenD
P/E TTM	12.17x	moomoo OpenD
52-Week Range	\$157.78–\$351.92	moomoo OpenD
Shares	280.8M	moomoo OpenD
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PART IV — COMPETITIVE LANDSCAPE

Competitor	Type	Mkt Cap	vs MPC
Valero Energy (VLO)	World's largest independent refiner	\$95.7B	VLO DGD SAF moat; MPC MPLX midstream moat
Phillips 66 (PSX)	Diversified downstream	\$90.9B	PSX CPCChem; MPC pure refining + midstream
PBF Energy (PBF)	Pure refiner	~\$5B	MPC 20x larger; MPLX moat PBF lacks
HF Sinclair (DINO)	Refiner + lubricants	~\$6B	Smaller regional; MPC scale advantage decisive

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Galveston Bay Crude Scheduler (COMPOSITE)

A MPC crude scheduler at Galveston Bay manages tanker deliveries of WCS heavy crude from the Trans Mountain Pipeline to the refinery dock. 'We're buying \$80 WCS crude and making \$105 worth of products. That's the Galveston Bay coker premium — our complex configuration processes what no one else can.' Represents MPC's largest single refinery advantage — Galveston Bay's ability to process discounted heavy crude that most refineries cannot handle.

COMPOSITE / ILLUSTRATIVE — drawn from US Gulf Coast complex refinery crude processing patterns

FINDING F-5-B: Portrait 2: The MPLX Distribution Investor (COMPOSITE)

An MLP investor holds MPLX for the 8.5% distribution yield. 'MPLX gives me infrastructure income — pipeline

fees, terminalling. Marathon keeps drilling into MPLX; my distributions grow every year. It's the boring part of MPC that pays for MPC's buybacks.' Represents MPLX's institutional income investor base — the stable yield that funds MPC's equity capital return programme.

COMPOSITE / ILLUSTRATIVE — drawn from US MLP midstream income investor patterns

FINDING F-5-C: Portrait 3: The MPC Share Buyback Beneficiary (COMPOSITE)

A long-term MPC investor who owned 1,000 shares in 2019 now represents 1.7% more of the company despite holding the same number. 'MPC has retired 40% of shares since 2019. My piece of MPC keeps getting larger without spending a dollar more.' Represents MPC's aggressive buyback programme — the compounding effect of share count reduction that increases EPS even without earnings growth.

COMPOSITE / ILLUSTRATIVE — drawn from MPC shareholder buyback benefit patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: Crack spread falls to \$10 — MPLX distributions sustain buybacks

EVIDENCE FOR:

Refinery crack spread falls to \$10; MPC refinery EBITDA falls 60%; but MPLX distributes \$3.5B to MPC regardless; MPC EPS falls from \$13 to \$6; buyback continues at slower pace; stock \$280-300

EVIDENCE AGAINST:

MPLX fee income is non-correlated with crack spread; MPC has used MPLX distributions to continue buybacks through prior crack trough

QUANTITATIVE TEST

$P(\text{MPC EPS} < \$7 \text{ in any fiscal year due to crack spread trough}) = 12\%$

VERDICT: BEAR — partial protection from MPLX; crack trough still impacts EPS significantly

Implication: Monitor crack spread; MPLX distribution coverage ratio; MPC quarterly FCF

HH2: HH2: MPC takes MPLX fully private — distribution disappears, buyback accelerates

EVIDENCE FOR:

MPC acquires MPLX minority at 10x EBITDA; owns 100% of midstream; eliminates distribution requirement to minority; all FCF goes to buybacks; EPS accretive \$1.50+; stock \$380+

EVIDENCE AGAINST:

MPC has historically preferred MPLX's public market access; minority buyout requires \$15B+ — significant leverage increase

QUANTITATIVE TEST

P(MPC takes MPLX fully private within 36 months)=15%

VERDICT: BULL — MPLX privatisation would be accretive but leveraged

Implication: Monitor MPC MPLX ownership %; any strategic review; MPLX unit price relative to MPC

HH3: HH3: MPC's share count reaches 200M by 2028 (from 280M current)

EVIDENCE FOR:

MPC continues \$8B+/year buybacks; share count falls from 280M to 200M by 2028; EPS mechanically rises 40% from count alone; stock \$450+

EVIDENCE AGAINST:

Buyback pace depends on FCF; crack trough would slow buybacks; MPLX distributions provide non-crack support for buyback

QUANTITATIVE TEST

P(MPC share count <220M by Q4 2028)=30%

VERDICT: BULL — share count reduction is MPC's most reliable compounding mechanism

Implication: Monitor MPC share count quarterly; buyback programme announcements; FCF-to-buyback ratio

HH4: HH4: EV adoption accelerates gasoline demand decline

EVIDENCE FOR:

US EV penetration reaches 30% of new sales by 2028; gasoline demand falls structurally 8%; MPC's non-complex gasoline-heavy refineries impaired

EVIDENCE AGAINST:

MPC's complex refineries can shift product slate from gasoline to diesel/jet; jet fuel growing; MPC Galveston Bay is flexible

QUANTITATIVE TEST

P(Structural gasoline demand decline forces MPC refinery reconfiguration >\$2B investment by 2028)=10%

VERDICT: BEAR — EV structural shift is slow but directional against MPC's gasoline volumes

Implication: Monitor US EV penetration; MPC gasoline vs diesel product mix quarterly

HH5: HH5: MPC announces renewable fuel conversion at a gasoline refinery

EVIDENCE FOR:

MPC converts a gasoline-heavy refinery (Anacortes, WA or Garyville) to renewable diesel/SAF; captures \$ 45Z credits + California LCFS; IRA windfall > refinery's gasoline earnings

EVIDENCE AGAINST:

MPC lacks Valero's DGD JV and Darling Ingredients feedstock supply; renewable fuel conversion requires feedstock infrastructure not yet in place

QUANTITATIVE TEST

P(MPC announces major renewable fuel conversion >50,000 B/day by 2027)=15%

VERDICT: BULL — IRA renewable fuel credit capture via refinery conversion is the logical next move

Implication: Monitor MPC renewable fuel strategy announcements; § 45Z regulatory guidance

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Share count <220M + crack spread recovery (P = 25%%)

MPC retires 60M shares; crack spread \$18+; EPS \$17+; stock \$400+

Driver: Share count quarterly; crack spread; MPLX distribution

Falsifier: Crack trough; buyback slows; share count falls only to 250M

SCENARIO 2: MPLX privatisation unlocks full midstream value (P = 12%%)

MPC takes MPLX private at \$50/unit; 100% midstream FCF; EPS accretive \$1.50; stock \$380+

Driver: MPC MPLX strategic review; unit price

Falsifier: Leverage too high; MPLX minority demands excessive premium

SCENARIO 3: Base case: crack stable, MPLX growing, stock \$330-360 (P = 40%%)

Crack spread \$15; MPLX distribution grows 5%/year; MPC shares fall to 255M by buyback; EPS \$13-15; stock steady

Driver: Quarterly EPS; crack spread; share count

Falsifier: Crack falls; buyback slows; stock \$280-300

SCENARIO 4: MPC renewable fuel conversion at Anacortes (P = 10%%)

Anacortes converted to renewable diesel; § 45Z credits add \$400M EBITDA; stock \$380+

Driver: MPC capital allocation; renewable fuel strategy

Falsifier: Feedstock supply constraints; MPC focuses on refining core

SCENARIO 5: Crack trough + MPLX stable (P = 8%%)

Crack \$10; MPLX \$3.5B distributions; EPS \$6; buyback continues at half pace; stock \$280-300; MPLX distributions prevent distress

Driver: Crack spread; MPLX distribution coverage

Falsifier: MPLX coverage ratio falls below 1.0x; distribution at risk

SCENARIO 6: Heavy crude arbitrage widens — Galveston Bay windfall (P = 15%%)

WCS-WTI differential widens to \$25; Galveston Bay captures \$3M/day extra margin; MPC EPS \$16+; stock \$380+

Driver: WCS-WTI differential; Trans Mountain Pipeline volumes

Falsifier: WCS-WTI narrows on Trans Mountain Pipeline opening; Galveston Bay margin compresses

SCENARIO 7: MPLX distribution increases 10%/year for 3 years (P = 20%%)

MPLX dropdown transactions from MPC; MPLX EBITDA grows; distributions to MPC reach \$5B; buyback accelerates further

Driver: MPLX EBITDA growth; distribution growth rate

Falsifier: Dropdown assets exhausted; MPLX organic growth insufficient for 10%/year

SCENARIO 8: MPC acquires PBF or HF Sinclair for refinery scale (P = 8%%)

MPC acquires small refiner at distressed valuation; adds complex capacity + MPLX dropdown potential; stock initially -5%, then +15% on synergies

Driver: PBF/DINO valuations; MPC M&A comments

Falsifier: MPC prefers buybacks over acquisitions; small refiner integration risk

SCENARIO 9: EV acceleration strands non-complex gasoline refinery (P = 8%%)

EV 30%+ new sales by 2028; MPC's simpler Midwestern refinery underperforms; MPC closes or sells 2 refineries; one-time charge; stock \$310-330

Driver: EV penetration data; gasoline demand quarterly

Falsifier: MPC complex refinery shift reduces gasoline exposure; Galveston Bay stays dominant

SCENARIO 10: Bull case: MPC = \$420+ by 2028 (P = 4%%)

Share count 200M + MPLX private + heavy crude premium + crack \$20 = \$420+

Driver: All upside catalysts simultaneously

Falsifier: Crack trough; EV headwind; MPLX minority expensive

PART VIII — SELDON VARIABLE DEEP DIVE

Variable	MPC Impact	Direction	Magnitude
Ω	Largest US refiner by capacity — fossil transformation node	↑ Increases	Ω-plus large
K	Refining infrastructure converts crude to industrial inputs for K	↑ Transitional	K+ transitional
ξ	Fuel distribution at scale makes energy accessible geographically	↑ Mild	ξ+ indirect
I	MPLX midstream infrastructure provides economic stability	↑ Moderate	I+ moderate

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
Crack spread trough	MEDIUM (20%)	HIGH	MPLX distributions fund buyback regardless of crack
EV structural shift	LOW (10%)	HIGH	Complex refinery product slate flexibility
MPLX distribution cut	VERY LOW (5%)	HIGH	1.3x distribution coverage; conservative MPLX management
Heavy crude WCS premium narrow	MEDIUM (15%)	MEDIUM	Trans Mountain Pipeline partially reduces WCS-WTI discount
M&A overpayment	LOW (10%)	MEDIUM	Management prefers buybacks; M&A historically disciplined

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: BEST-IN-CLASS REFINER + MIDSTREAM — MPLX FUNDS
BUYBACK MACHINE

MPC at \$350.875 (mkt cap \$98.534B, P/E 12.17x, EPS \$13.22) is the most complete downstream energy company. MPLX distributions fund buybacks through crack cycles. Galveston Bay complex refinery captures heavy crude premium. 40%+ share buyback over 5 years is the most aggressive in the sector. Seldon: Ω -plus (refining); MPLX infrastructure provides I+ stability.

HUMAN_ONLY_MANUAL. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,555ch.